A.E.G.I.S. MK2

Wiring guide

15 steps, each with its own drawing and a check. Wire as you assemble, in this order:

BENCH	Solder headers on the modules, build the ESP32-S3 carrier, the fuse board, the switch leads and the power bus (W1-W3). Easier on the bench than inside the rover.
ASSEMBLY 1-15	Motors, wheels, edge sensors, intake bay, battery, lower-deck boards (Assembly guide).
STEP 16: WIRE	Lower deck: power, set 5.0 V, motors, servo power, IR sensors, then label the leads for the upper deck (W4-W9).
ASSEMBLY 17-23	Pillars, upper deck, switch, carrier with the ESP32-S3 removed, gyro, mics, side sensors.
STEP 24: WIRE	Upper deck: power, I2C, microphones, motor control, encoders and servo signal (W10-W14).
POWER-UP	First power-up checks (W15), then assembly 25-27 and the Deployment guide.

Rules for every wire

1. **Battery unplugged** whenever you solder or move a wire.
2. Cut each wire 3 cm longer than the route, strip 5 mm, twist the strands, tin them.
3. Slide heat-shrink on **before** soldering; shrink it over every power joint.
4. Tug test: pull each finished joint gently. It must not move.
5. Red = plus, black = ground, always. Signal wires: white or green, flagged with masking tape.
6. One row of the table at a time; tick it off on a printed copy.
7. Never connect a regulator output before it reads 5.0 V (W5).

Wire	Use
18 AWG red / black	battery, fuse, switch, power bus, regulator and driver inputs
22 AWG 3-core	5 V outputs, servo power, motor M1 / M2
26 AWG red / black	3.3 V and ground to sensors
26 AWG white / green	all signals

Pin numbers match the firmware (aegis_config.h). The reference pages at the end show every board face and the full pin map. Clone boards sometimes order their pins differently: **always read the label printed on the board**.

W1 Before assembly: solder headers on the modules

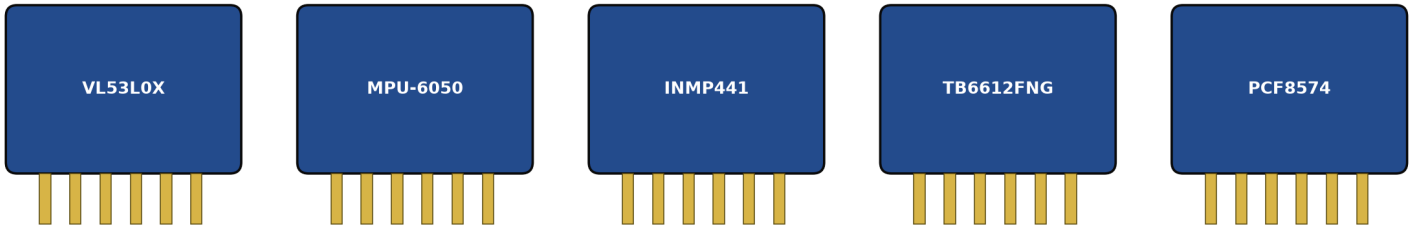
Bench, before assembly

YOU NEED

VL53L0X, MPU-6050, INMP441, TB6612 headers

solder

Short pins down, long pins up. Push the header into a breadboard or a scrap of perfboard to hold it square, lay the module on top, solder one corner pin, check it is square, then solder the rest.



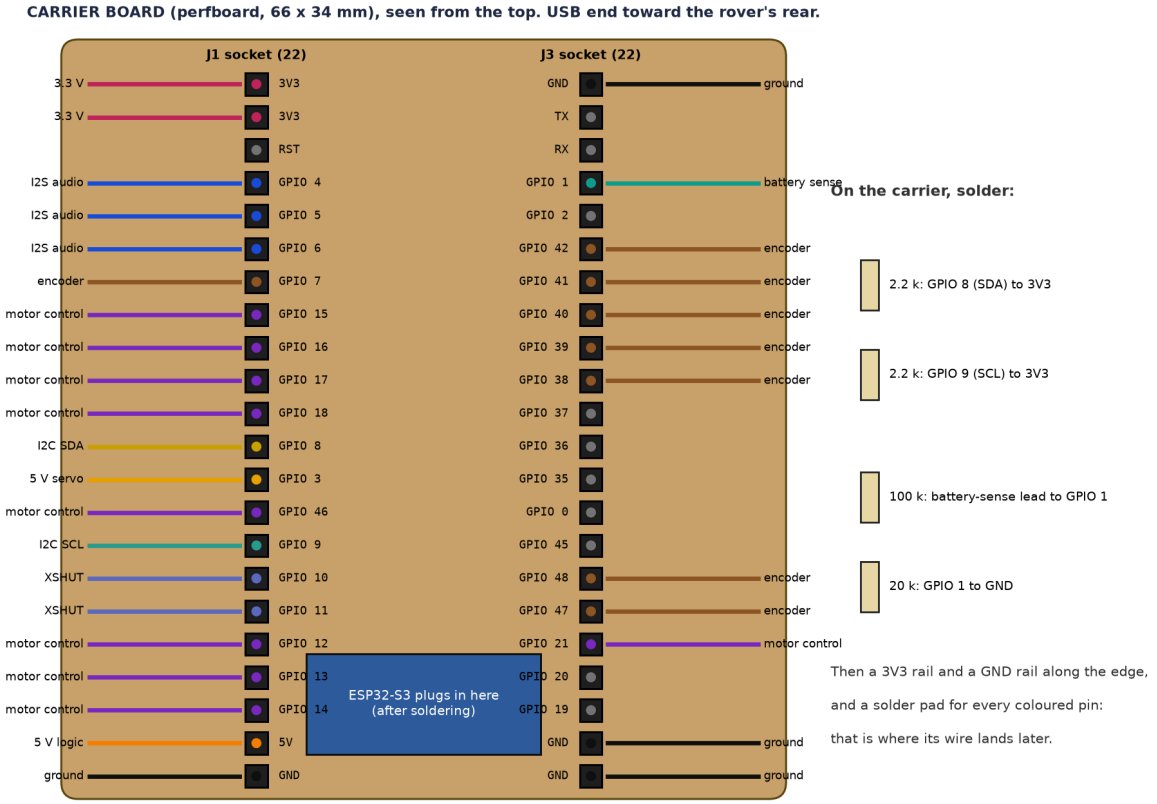
6 pins	8 pins	6 pins	16 pins	if loose
VL53L0X (x3)	MPU-6050	INMP441 (x2)	TB6612FNG (x2)	PCF8574

Each joint: heat pin and pad together 2 s, feed solder until it flows round the pin, remove solder, then iron. A good joint is a small shiny cone.

CHECK BEFORE YOU GO ON. Each module sits flat and square on its header; no solder bridges between pins (check with a magnifier).

W2 Before assembly: build the ESP32-S3 carrier board

Bench, before assembly



Sockets: cut each 22-way row from one full 1x20 strip plus 2 sockets from a spare strip. The two rows are 22.86 mm (9 holes) apart: check with the ESP32-S3 itself before soldering.

CHECK BEFORE YOU GO ON. Meter on resistance: GPIO 8 to 3V3 reads 2.2 k, GPIO 9 to 3V3 reads 2.2 k, GPIO 1 to GND reads 20 k.

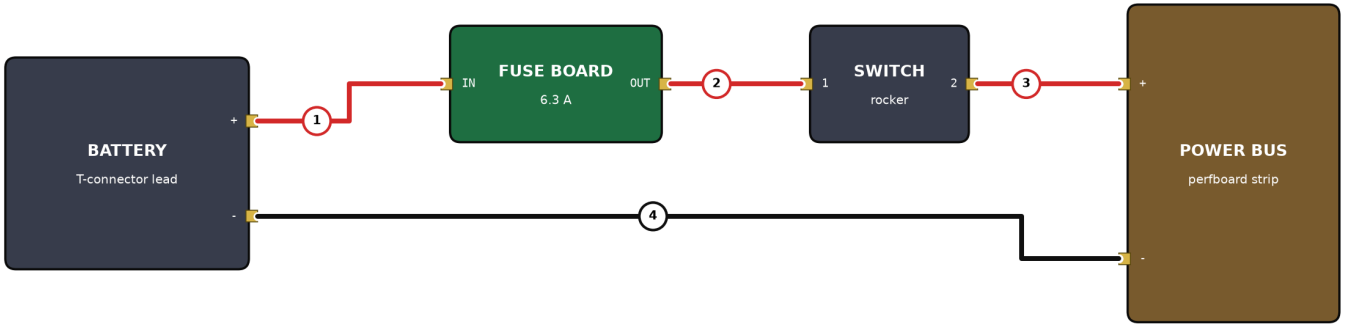
YOU NEED

2.2 k resistors x 2, 100 k x 1, 20 k x 1
resistor kit item #34

1x20 female strips
4 item #32

W3 Before assembly: battery lead, fuse, switch, power bus

Bench, before assembly



Make the POWER BUS from a 2 x 8 strip of perfboard: one row for +, one row for -, each row bridged with solder. Everything that needs battery power connects to it.

#	From	To	Wire
1	BATTERY +	FUSE BOARD IN	18 AWG red
2	FUSE BOARD OUT	SWITCH 1	18 AWG red

#	From	To	Wire
3	SWITCH 2	POWER BUS +	18 AWG red
4	BATTERY -	POWER BUS -	18 AWG black

CHECK BEFORE YOU GO ON. Switch OFF: no continuity from battery + to the bus +. Switch ON: continuity. Never any continuity from + to -.

YOU NEED

18 AWG red, 3 pieces x 15 cm
item #24

18 AWG black, 1 x 25 cm
item #25

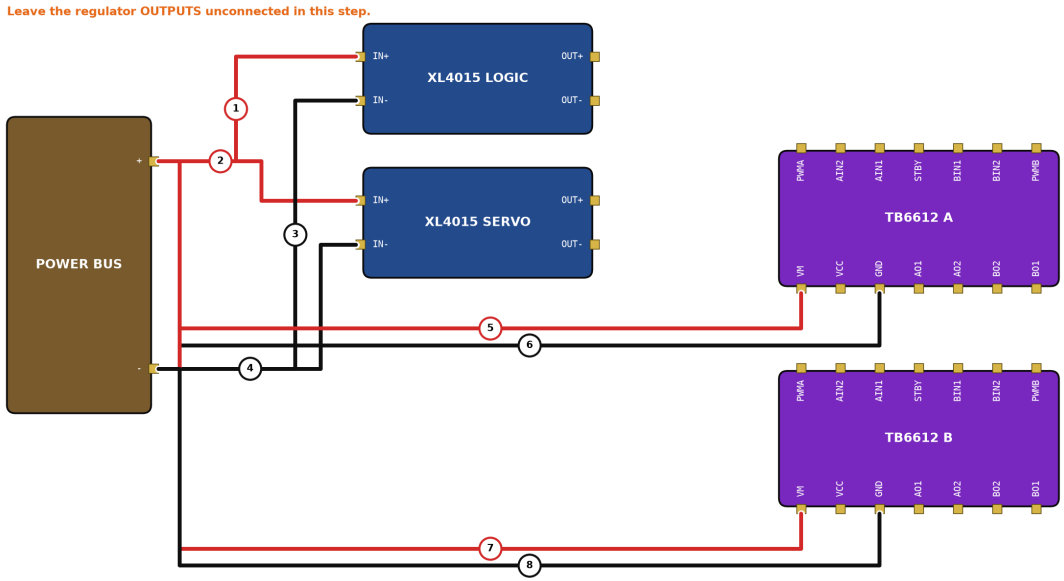
Heat-shrink
item #35

WIRE COLOURS

- battery +: 18 AWG red
- ground: 18 AWG black (power) / 26 AWG black (signal)

W4 Step 16 a: power to regulators and motor drivers

Assembly step 16: lower deck



#	From	To	Wire
1	POWER BUS +	XL4015 LOGIC IN+	18 AWG red
2	POWER BUS +	XL4015 SERVO IN+	18 AWG red
3	POWER BUS -	XL4015 LOGIC IN-	18 AWG black
4	POWER BUS -	XL4015 SERVO IN-	18 AWG black

#	From	To	Wire
5	POWER BUS +	TB6612 A VM	18 AWG red
6	POWER BUS -	TB6612 A GND	18 AWG black
7	POWER BUS +	TB6612 B VM	18 AWG red
8	POWER BUS -	TB6612 B GND	18 AWG black

CHECK BEFORE YOU GO ON. Switch OFF, battery unplugged: no continuity between + and - anywhere.

YOU NEED

18 AWG red / black, about 15 cm each

8 wires item #24, #25

WIRE COLOURS

- battery +: 18 AWG red
- ground: 18 AWG black (power) / 26 AWG black (signal)

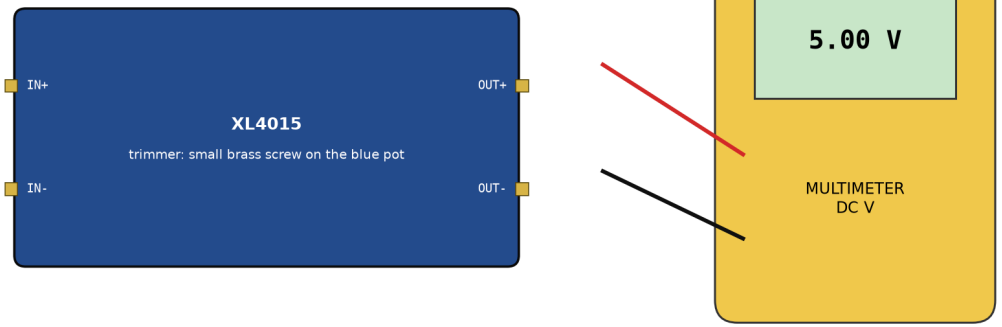
W5 Step 16 b: set both regulators to 5.0 V

Assembly step 16: lower deck

YOU NEED

- Multimeter
  lab

1. Battery plugged in, switch ON, nothing connected to either regulator's output.
2. Meter on DC volts: red probe on OUT+, black on OUT-.
3. Turn the trimmer (often 10+ turns) until it reads 5.0 V. Do the same for the second regulator.

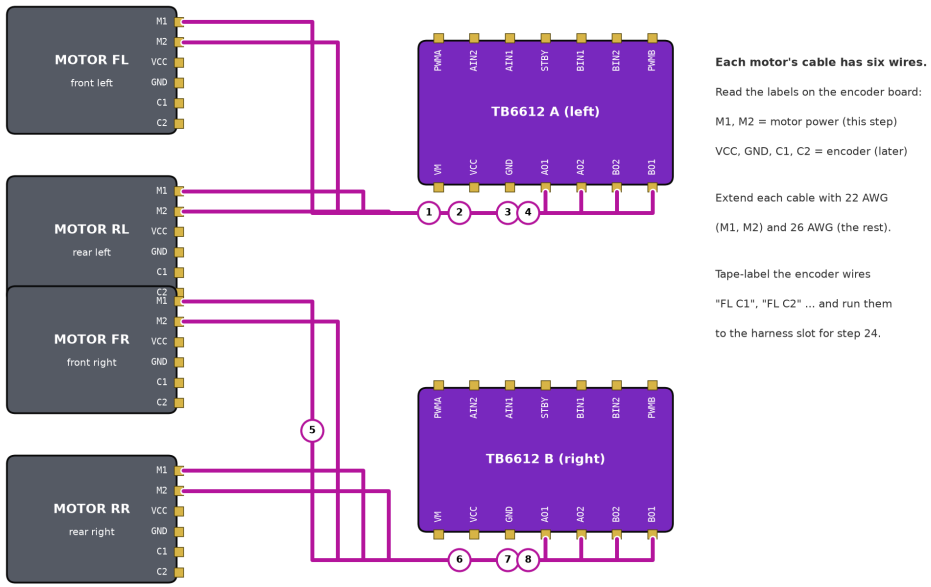


4. Switch OFF and unplug the battery before the next step.

CHECK BEFORE YOU GO ON. Both regulators read 4.95-5.05 V with nothing connected to their outputs.

W6 Step 16 c: motors to the motor drivers

Assembly step 16: lower deck



Each motor's cable has six wires.

Read the labels on the encoder board:

M1, M2 = motor power (this step)

VCC, GND, C1, C2 = encoder (later)

Extend each cable with 22 AWG

(M1, M2) and 26 AWG (the rest).

Tape-label the encoder wires

"FL C1", "FL C2" ... and run them

to the harness slot for step 24.

#	From	To	Wire
1	MOTOR FL M1	TB6612 A (left) AO1	22 AWG
2	MOTOR FL M2	TB6612 A (left) AO2	22 AWG
3	MOTOR RL M1	TB6612 A (left) BO1	22 AWG
4	MOTOR RL M2	TB6612 A (left) BO2	22 AWG

#	From	To	Wire
5	MOTOR FR M1	TB6612 B (right) AO1	22 AWG
6	MOTOR FR M2	TB6612 B (right) AO2	22 AWG
7	MOTOR RR M1	TB6612 B (right) BO1	22 AWG
8	MOTOR RR M2	TB6612 B (right) BO2	22 AWG

CHECK BEFORE YOU GO ON. Each motor: M1 to M2 reads a few ohms through the motor.

YOU NEED

22 AWG 3-core, 4 x 20 cm
item #26

26 AWG for encoder extensions
item #27-#30

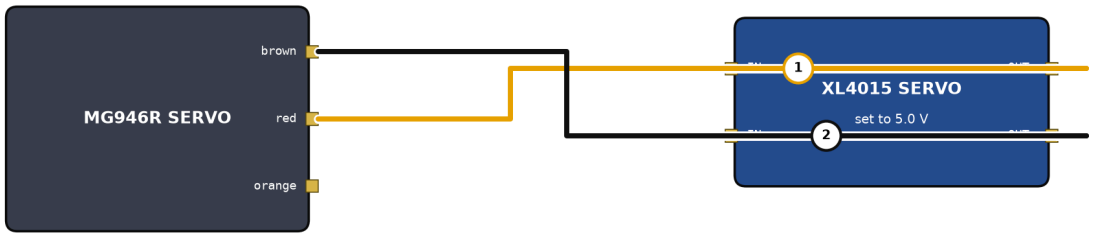
WIRE COLOURS

motor power: 22 AWG (3-core) red / black

W7 Step 16 d: servo power

Assembly step 16: lower deck

Cut the servo plug off, strip 5 mm, solder to the SERVO regulator's output, heat-shrink each joint.



Orange = signal: extend it with 26 AWG, label it "SERVO" and leave it at the harness slot for step 24.

#	From	To	Wire
1	MG946R SERVO red	XL4015 SERVO OUT+	22 AWG red

#	From	To	Wire
2	MG946R SERVO brown	XL4015 SERVO OUT-	18 AWG black

CHECK BEFORE YOU GO ON. Servo red to brown: no short (more than 1 k).

YOU NEED

22 AWG red / black, 2 x 15 cm

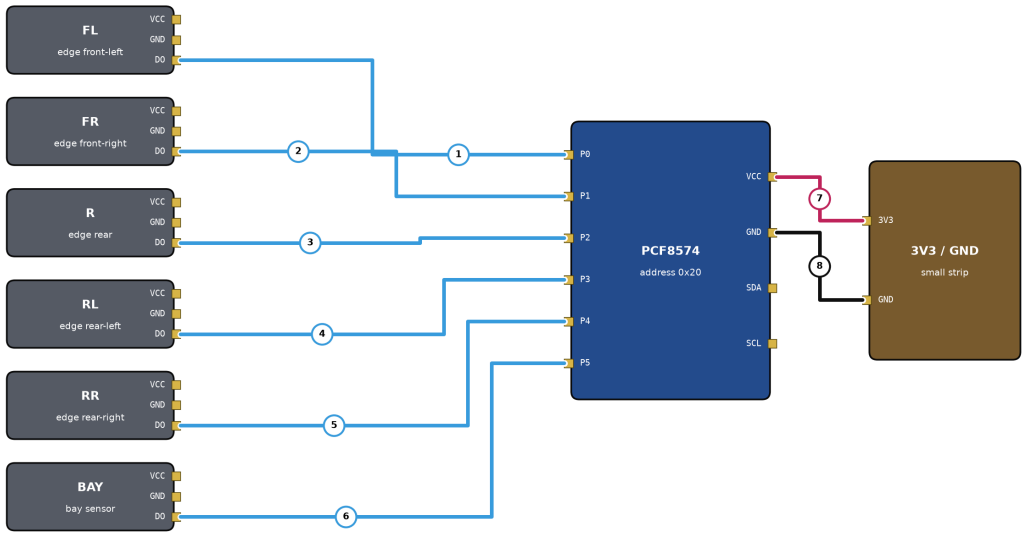
item #26

WIRE COLOURS

- 5 V servo: 22 AWG red (3-core)
- ground: 18 AWG black (power) / 26 AWG black (signal)

W8 Step 16 e: edge sensors and bay sensor to the PCF8574

Assembly step 16: lower deck



Daisy-chain VCC and GND of all six sensors to the 3V3 / GND strip (not drawn). PCF8574 SDA / SCL and the strip's 3V3 lead go up to the carrier in step 24: label them.

#	From	To	Wire
1	FL DO	PCF8574 P0	26 AWG white
2	FR DO	PCF8574 P1	26 AWG white
3	R DO	PCF8574 P2	26 AWG white
4	RL DO	PCF8574 P3	26 AWG white

#	From	To	Wire
5	RR DO	PCF8574 P4	26 AWG white
6	BAY DO	PCF8574 P5	26 AWG white
7	PCF8574 VCC	3V3 / GND 3V3	26 AWG red
8	PCF8574 GND	3V3 / GND GND	18 AWG black

CHECK BEFORE YOU GO ON. Each sensor's DO has continuity to its PCF8574 pin P0-P5 and to nothing else.

YOU NEED

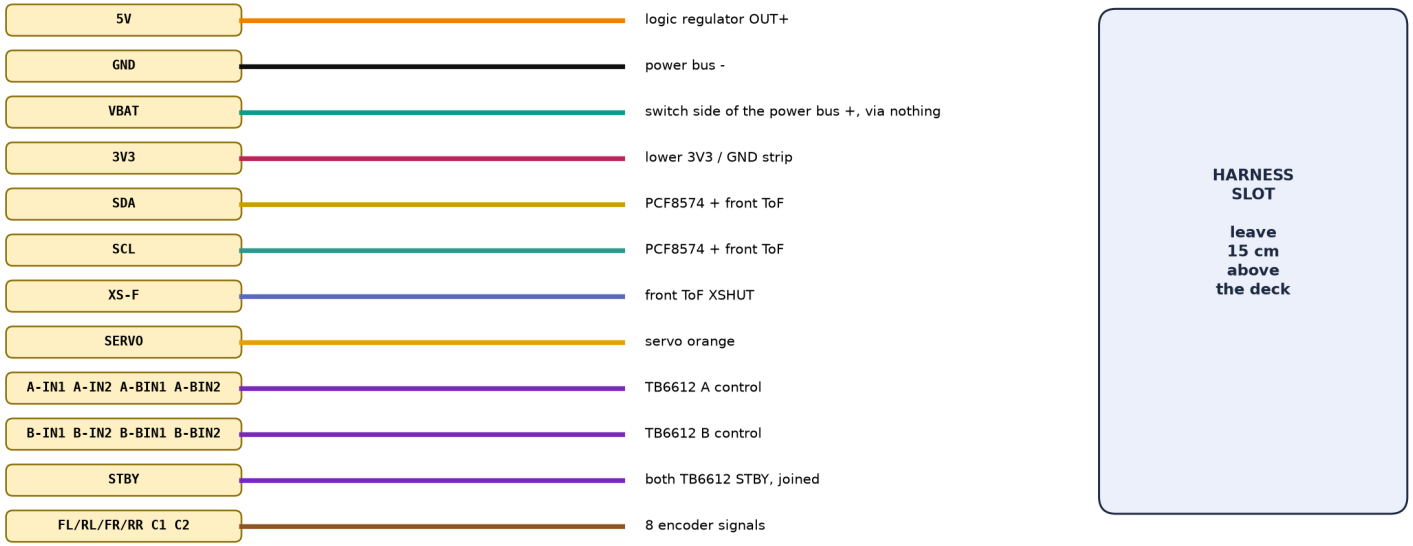
26 AWG white (signals), red / black (3V3, GND)
about 4 m item #27-#30

WIRE COLOURS

- IR signal: 26 AWG white
- 3.3 V: 26 AWG red
- ground: 18 AWG black (power) / 26 AWG black (signal)

W9 Step 16 f: label and route the leads for the upper deck

Assembly step 16: lower deck



Masking-tape flag on every lead with the tag shown. Encoder VCC/GND and TB6612 VCC go to the 3V3 / GND strip; tie TB6612 PWMA and PWMB to its own VCC with a short wire.

CHECK BEFORE YOU GO ON. Every lead has a tag and reaches 15 cm above the deck through the harness slot.

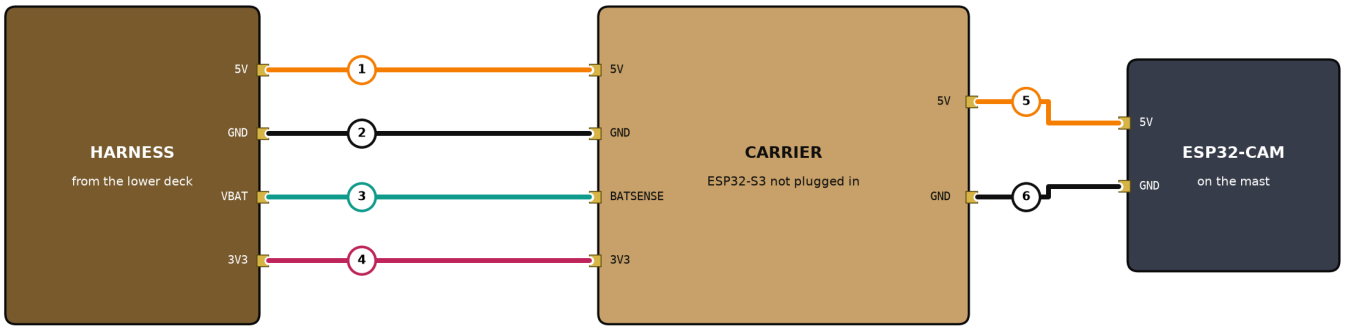
YOU NEED

Masking tape, pen
your own

W10 Step 24 a: power to the carrier and the camera

Assembly step 24: upper deck

BATSENSE is the top of the 100 k / 20 k divider on the carrier.



#	From	To	Wire
1	HARNESS 5V	CARRIER 5V	22 AWG red
2	HARNESS GND	CARRIER GND	18 AWG black
3	HARNESS VBAT	CARRIER BATSENSE	26 AWG white

#	From	To	Wire
4	HARNESS 3V3	CARRIER 3V3	26 AWG red
5	CARRIER 5V	ESP32-CAM 5V	22 AWG red
6	CARRIER GND	ESP32-CAM GND	18 AWG black

CHECK BEFORE YOU GO ON. Carrier 5V to GND: no short. BATSENSE to GND reads about 20 k.

YOU NEED

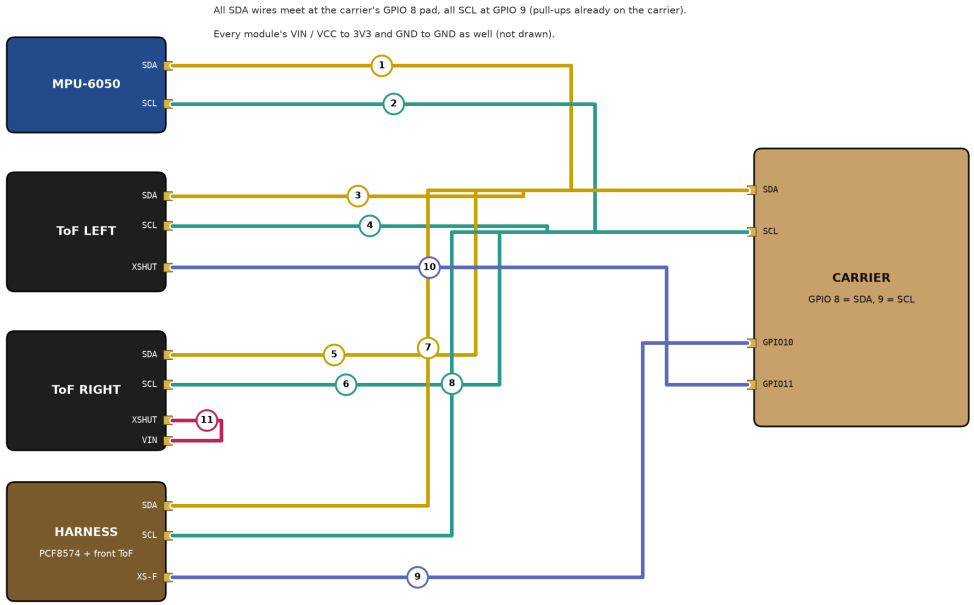
22 AWG red / black
item #26

WIRE COLOURS

- 5 V logic: 22 AWG red (3-core)
- ground: 18 AWG black (power) / 26 AWG black (signal)
- battery sense: 26 AWG white
- 3.3 V: 26 AWG red

W11 Step 24 b: I2C sensors (gyro, distance sensors, PCF8574)

Assembly step 24: upper deck



#	From	To	Wire
1	MPU-6050 SDA	CARRIER SDA	26 AWG white
2	MPU-6050 SCL	CARRIER SCL	26 AWG green
3	ToF LEFT SDA	CARRIER SDA	26 AWG white
4	ToF LEFT SCL	CARRIER SCL	26 AWG green
5	ToF RIGHT SDA	CARRIER SDA	26 AWG white

#	From	To	Wire
7	HARNESS SDA	CARRIER SDA	26 AWG white
8	HARNESS SCL	CARRIER SCL	26 AWG green
9	HARNESS XS-F	CARRIER GPIO10	26 AWG green
10	ToF LEFT XSHUT	CARRIER GPIO11	26 AWG green
11	ToF RIGHT XSHUT	ToF RIGHT VIN	26 AWG red

CHECK BEFORE YOU GO ON. GPIO 8 has continuity to every SDA pin; GPIO 9 to every SCL pin; SDA and SCL are not connected to each other.

YOU NEED

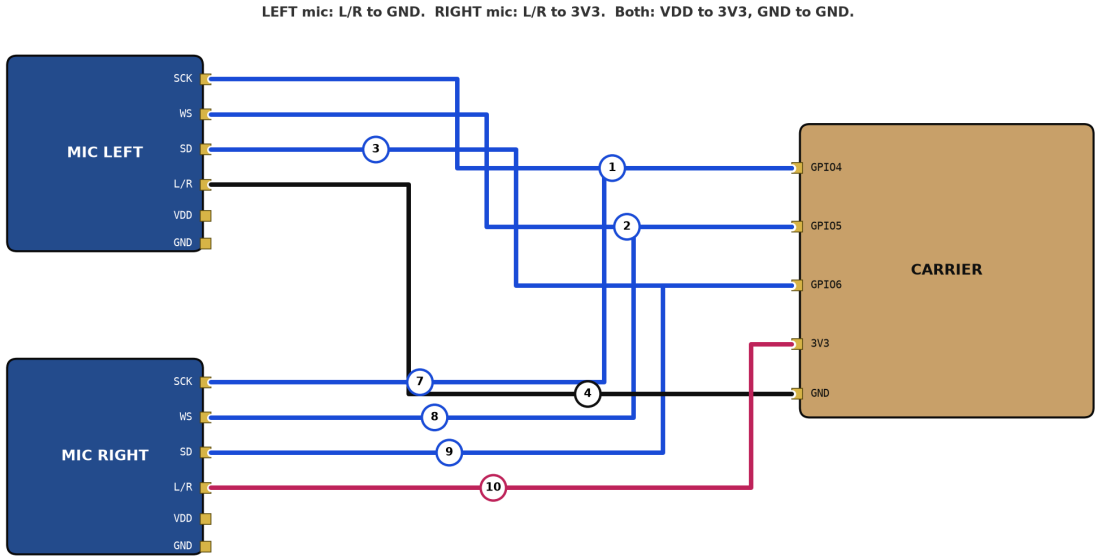
26 AWG white (SDA), green (SCL), red / black (3V3, GND)
item #27-#30

WIRE COLOURS

- I2C SDA: 26 AWG white
- I2C SCL: 26 AWG green
- XSHUT: 26 AWG green
- 3.3 V: 26 AWG red

W12 Step 24 c: microphones

Assembly step 24: upper deck



Keep these wires under 15 cm and away from the motor leads: audio is the most noise-sensitive signal.

#	From	To	Wire
1	MIC LEFT SCK	CARRIER GPIO4	26 AWG white
2	MIC LEFT WS	CARRIER GPIO5	26 AWG white
3	MIC LEFT SD	CARRIER GPIO6	26 AWG white
4	MIC LEFT L/R	CARRIER GND	18 AWG black

#	From	To	Wire
7	MIC RIGHT SCK	CARRIER GPIO4	26 AWG white
8	MIC RIGHT WS	CARRIER GPIO5	26 AWG white
9	MIC RIGHT SD	CARRIER GPIO6	26 AWG white
10	MIC RIGHT L/R	CARRIER 3V3	26 AWG red

CHECK BEFORE YOU GO ON. Left mic L/R to GND: continuity. Right mic L/R to 3V3: continuity.

YOU NEED

26 AWG, under 15 cm each

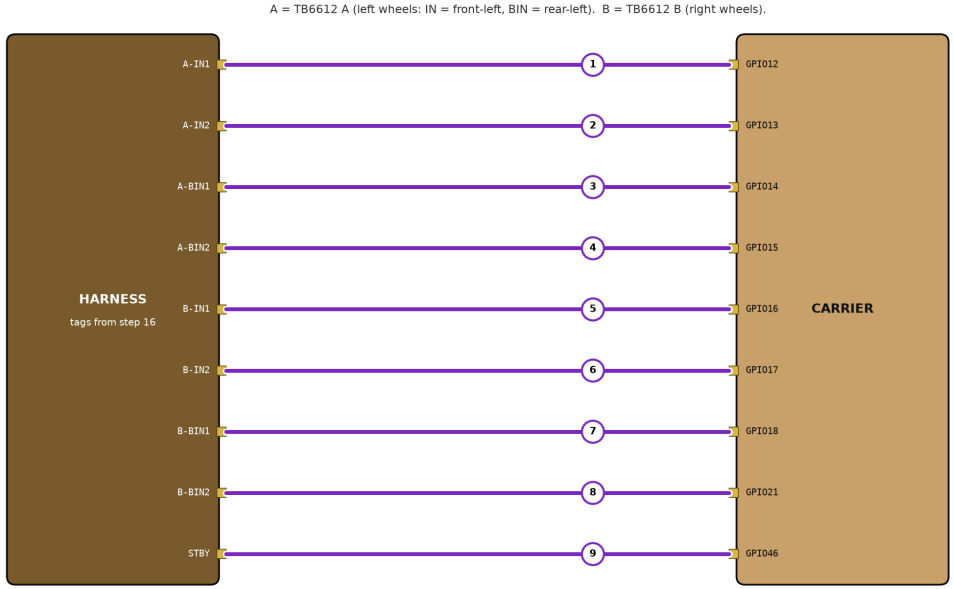
12 wires item #27-#30

WIRE COLOURS

- I2S audio: 26 AWG white
- ground: 18 AWG black (power) / 26 AWG black (signal)
- 3.3 V: 26 AWG red

W13 Step 24 d: motor-driver control

Assembly step 24: upper deck



#	From	To	Wire
1	HARNESS A-IN1	CARRIER GPIO12	26 AWG white
2	HARNESS A-IN2	CARRIER GPIO13	26 AWG white
3	HARNESS A-BIN1	CARRIER GPIO14	26 AWG white
4	HARNESS A-BIN2	CARRIER GPIO15	26 AWG white
5	HARNESS B-IN1	CARRIER GPIO16	26 AWG white

#	From	To	Wire
6	HARNESS B-IN2	CARRIER GPIO17	26 AWG white
7	HARNESS B-BIN1	CARRIER GPIO18	26 AWG white
8	HARNESS B-BIN2	CARRIER GPIO21	26 AWG white
9	HARNESS STBY	CARRIER GPIO46	26 AWG white

CHECK BEFORE YOU GO ON. Each tagged lead has continuity to its GPIO pad and to no neighbour.

YOU NEED

Harness leads from step 16

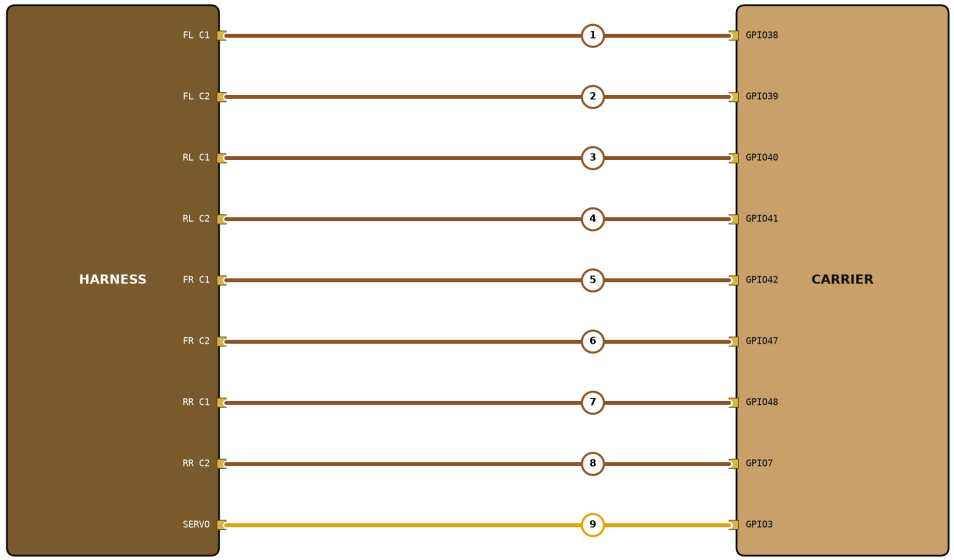
9

WIRE COLOURS

motor control: 26 AWG white

W14 Step 24 e: encoders and servo signal

Assembly step 24: upper deck



#	From	To	Wire
1	HARNESS FL C1	CARRIER GPIO38	26 AWG green
2	HARNESS FL C2	CARRIER GPIO39	26 AWG green
3	HARNESS RL C1	CARRIER GPIO40	26 AWG green
4	HARNESS RL C2	CARRIER GPIO41	26 AWG green
5	HARNESS FR C1	CARRIER GPIO42	26 AWG green

#	From	To	Wire
6	HARNESS FR C2	CARRIER GPIO47	26 AWG green
7	HARNESS RR C1	CARRIER GPIO48	26 AWG green
8	HARNESS RR C2	CARRIER GPIO7	26 AWG green
9	HARNESS SERVO	CARRIER GPIO3	22 AWG red

CHECK BEFORE YOU GO ON. Each tagged lead has continuity to its GPIO pad and to no neighbour.

YOU NEED

Harness leads from step 16

9

WIRE COLOURS

- encoder: 26 AWG green
- 5 V servo: 22 AWG red (3-core)

W15 First power-up

Assembly step 24: upper deck

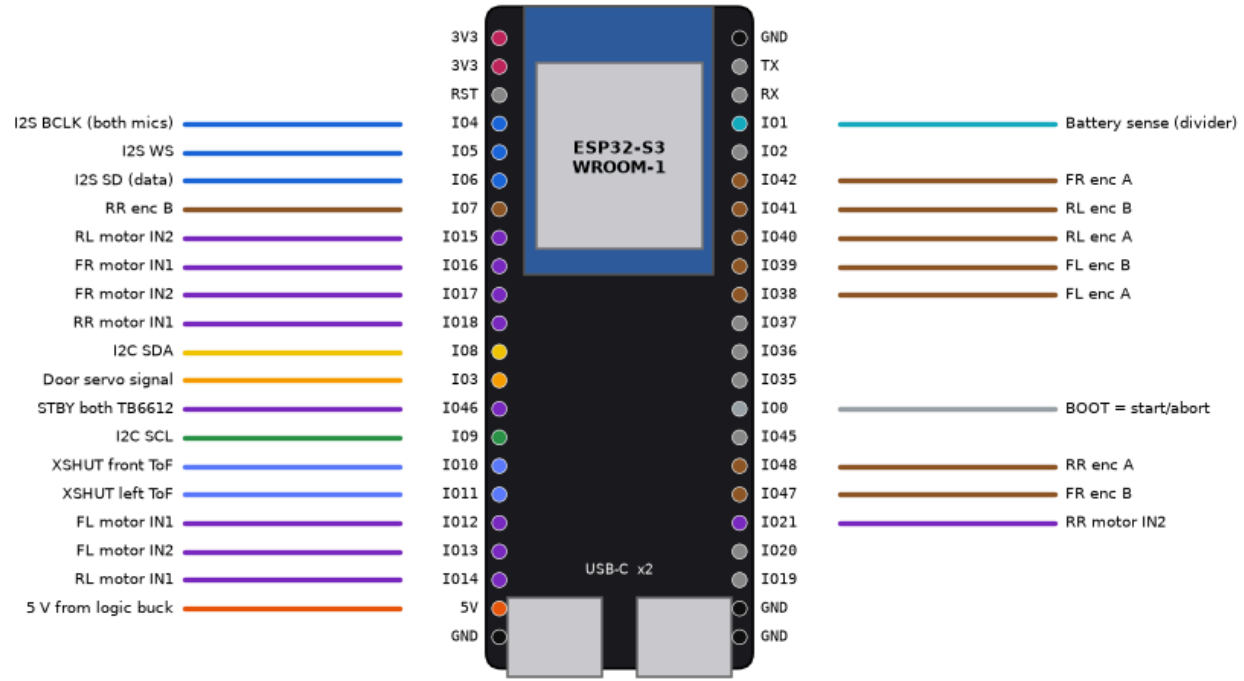
YOU NEED

- Multimeter
  lab

1. ESP32-S3 OUT of its sockets. Switch OFF. Battery plugged in.
2. Meter on continuity: carrier 5V to GND, 3V3 to GND, battery + to GND: no beep on any.
3. Switch ON. Carrier 5V pad reads 4.9-5.1 V. Servo regulator output reads 4.9-5.1 V.
4. Switch OFF. Plug the ESP32-S3 in, USB toward the rear. Switch ON: its red LED lights.
5. Carrier 3V3 pad now reads 3.2-3.4 V (from the ESP32-S3's regulator).
6. Flash and run bring-up: AEGIS_Deployment_Guide.pdf, sections 2-5.

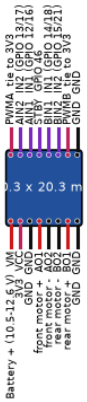
CHECK BEFORE YOU GO ON. If any reading is wrong: switch OFF and find it before plugging the ESP32-S3 in.

A.E.G.I.S. wiring guide 1/3: ESP32-S3-DevKitC-1 pin assignment (top view, antenna up)

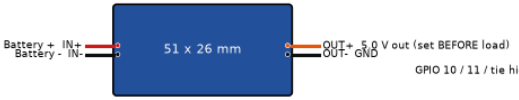


A.E.G.I.S. wiring guide 2/3: module faces, pins and where each wire lands (top views, to scale)

TB6612FNG (x2: A = left wheels, B = right)



XL4015 buck (x2: logic 5.0 V, servo 5.0 V)



GY-530 VL53L0X (x3: front, left, right)



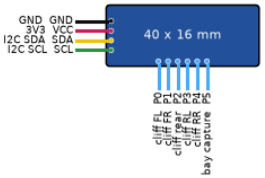
GY-521 MPU-6050



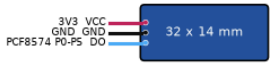
INMP441 mic (x2)



PCF8574T I/O expander



TCRT5000 x5 / IR obstacle module x1



MG946R servo | N20 encoder motor (x4)



Battery 12 V
Ground
5 V logic
5 V servo
3.3 V
I2C SDA
I2C SCL
I2S audio
Motor PWM / STBY
Encoders
ToF XSHUT
IR sensors
UART / misc
Battery sense

A.E.G.I.S. wiring guide 3/3: wire list

From	To	Signal	Wire
Battery + (T-connector lead)	6.3 A fuse, power switch, then both XL4015 IN+ and both TB6612 VM	10.5-12.6 V	18 AWG
Battery -	Both XL4015 IN-, all GND	0 V	18 AWG
XL4015 logic OUT+	ESP32-S3 5V, ESP32-CAM 5V	5.0 V	22 AWG
XL4015 servo OUT+	MG946R red	5.0 V	22 AWG
ESP32-S3 3V3	sensors VCC, TB6612 VCC + PWMA/PWMB, encoders	3.3 V	26 AWG
ESP32 GPIO 8 / 9	VL53L0X x3, MPU-6050, PCF8574 (SDA/SCL), 2.2 k pull-ups	I2C 400 kHz	26 AWG
ESP32 GPIO 4 / 5 / 6	both INMP441 SCK / WS / SD	I2S 48 kHz	26 AWG, short
ESP32 GPIO 10 / 11	front / left VL53L0X XSHUT (right: tie to VIN)	address select	26 AWG
ESP32 GPIO 12-18, 21	TB6612 AIN1/AIN2/BIN1/BIN2	20 kHz PWM	26 AWG
ESP32 GPIO 46	STBY on both TB6612	enable	26 AWG
TB6612 AO1/AO2/BO1/BO2	motor red / white	motor power	22 AWG
Encoders yellow / green	GPIO 38/39, 40/41, 42/47, 48/7	quadrature	26 AWG
ESP32 GPIO 3	MG946R orange	50 Hz servo pulse	26 AWG
Battery + via 100k / 20k	ESP32 GPIO 1 (divider tapped after the switch)	battery voltage	26 AWG
IR modules DO	PCF8574 P0-P5	5 cliff + bay capture	26 AWG

The fuse and the power switch sit in the battery + line before anything else. The 2.2 k pull-ups and the battery divider are on the ESP32-S3 carrier board.
Common ground star point at the battery negative. Set both XL4015 outputs to 5.0 V with a meter before connecting any board. Twist each motor's two leads together. Keep the I2S wires short (< 15 cm).
Header orders follow the reference boards; clone modules sometimes differ: check the silkscreen before soldering.